Build Weather — User Guide

Version 0.4

Build Weather turns a C++ build into a picture of your source tree. Instead of reading a scrolling wall of compiler output, you look at a treemap: every file is a rectangle, its area is what it costs to build, and its colour is how long it took. Watching a build run looks like weather crossing a map. Watching one afterwards tells you which header is quietly adding minutes to every rebuild.

This guide walks through each part of the window, then through the four things you will actually do with it.

1. Before you start

Build Weather reads three files that your build system already produces. You do not have to instrument anything.

FILE	WHERE IT COMES FROM	WHAT IT GIVES YOU
`.ninja_log`	Written by ninja automatically	Exact start and end time of every build step
`compile_commands.json`	CMake's `EXPORT_COMPILE_COMMANDS` option	Which source file each object came from
`*.json` next to each object	clang-cl with `-ftime-trace`	Per-header and per-template cost inside each file

Only the first is required. The second makes the map show your source files instead of paths under `CMakeFiles/`. The third unlocks the header and template rankings on the Analysis tab.

So: **configure your project with CMake and the Ninja generator**, build it once, and point Build Weather at the build directory.

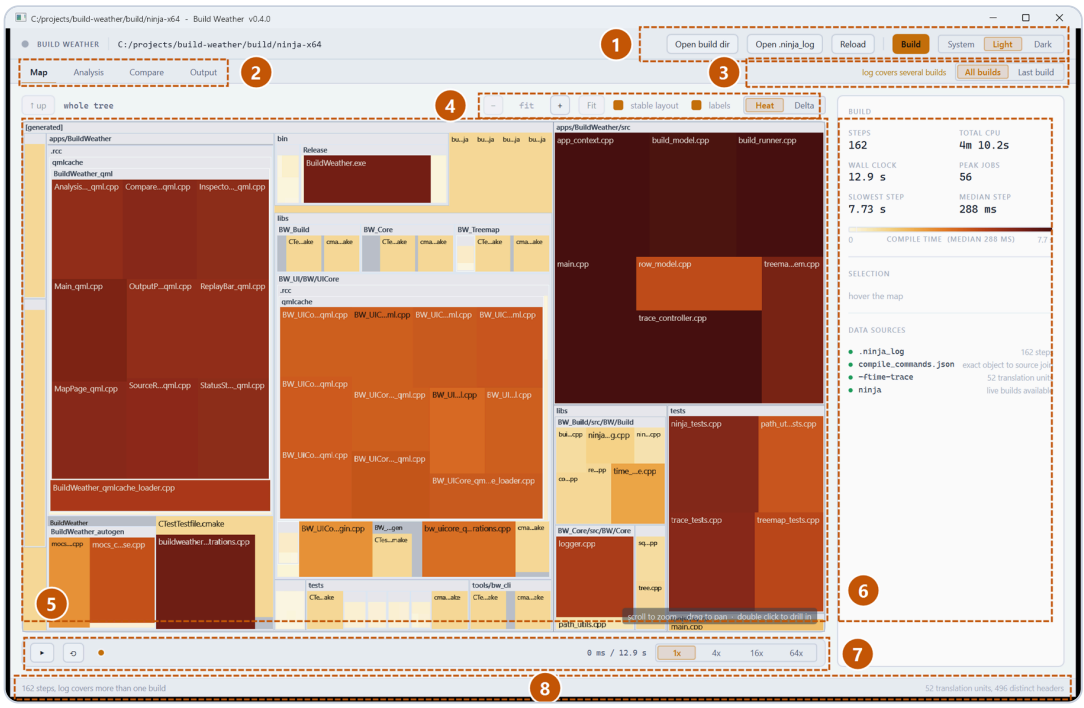
```
cmake -S . -B build/ninja -G Ninja -DCMAKE_EXPORT_COMPILE_COMMANDS=ON
cmake --build build/ninja
```

> **Live builds need a developer environment.** `cl.exe` cannot find the standard headers unless `INCLUDE` and `LIB` are set, and an application started from Explorer does not have them. Start Build Weather from a Developer Command Prompt if you want to run builds from inside it. The app checks this and warns you on the Output tab rather than letting the map fill up with identical errors.

2. The window

Open a build directory with **Open build dir**, or pass it on the command line:

```
BuildWeather.exe build\ninja --source . --traces build\clangcl
```

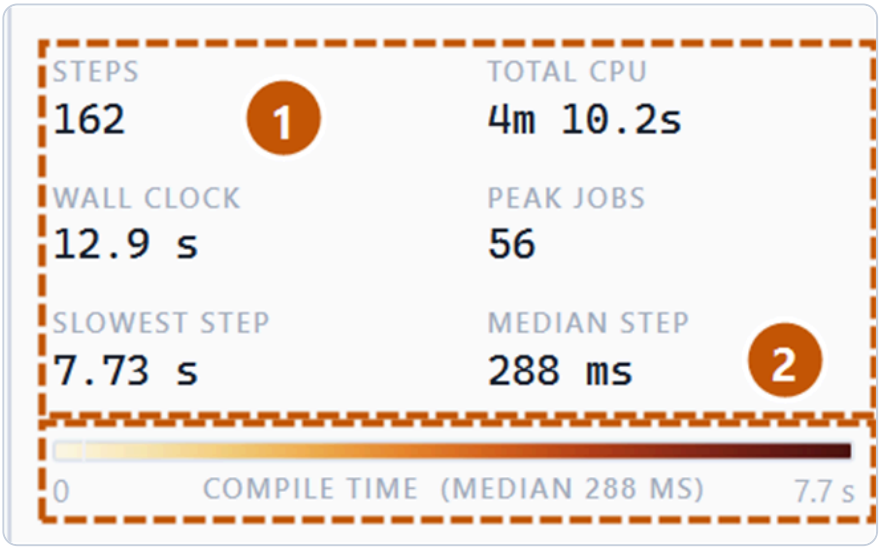


The Map tab, with the main regions marked

1. **File actions.** Open a build directory or a `.ninja_log` directly, re-read it after another build, and start a build from inside the app. The theme switch on the right follows your system by default.
2. **Tabs.** Map, Analysis, Compare and Output. The Map is the picture; the Analysis tab is where the numbers live.
3. **Scope.** A `.ninja_log` accumulates across every build you have ever run in that directory. *All builds* takes each target's most recent entry, which answers "what does a full build of this tree cost". *Last build* keeps only the most recent ninja run. The note beside it appears when the log covers more than one build, because that changes what the timeline means.
4. **Map controls.** Zoom out, current zoom, zoom in, Fit. Then *stable layout* and *labels*, and the colour mode: **Heat** for compile time, **Delta** for the difference against a baseline build.
5. **The map.** Directories nest, each file is a leaf. Area is compile time, colour is compile time on the scale shown in the panel to the right.
6. **Build summary and legend.** Totals, the colour scale with the median marked on it, whatever you last hovered or clicked, and which data sources were found.
7. **Replay transport.** Scrub through the build and watch it happen again.
8. **Status.** What is loaded, and during a build the progress, elapsed time and how many compiler jobs are actually running.

3. Reading the map

Every rectangle is one build step. **Area is compile time** and **colour is compile time**, on the scale shown in the summary panel.



The build summary and the colour scale

1. **Totals.** Steps in the snapshot, the CPU time they add up to, the wall clock the build actually took, the slowest single step and the median one. The gap between median and slowest is the shape of your problem: a high median means the whole build is slow, while a median of 200 ms next to a slowest step of 90 s means a handful of files are the whole story.
2. **The colour scale**, with the median marked on it as a tick.

The scale flips with the theme so that "expensive" is always the colour furthest from the page: pale cream to deep red on the light theme, near-black to bright yellow on the dark one. Encoding time twice, as area and as colour, is deliberate: size finds the expensive files from across the room, and colour still works for the small cells where area has run out of pixels.

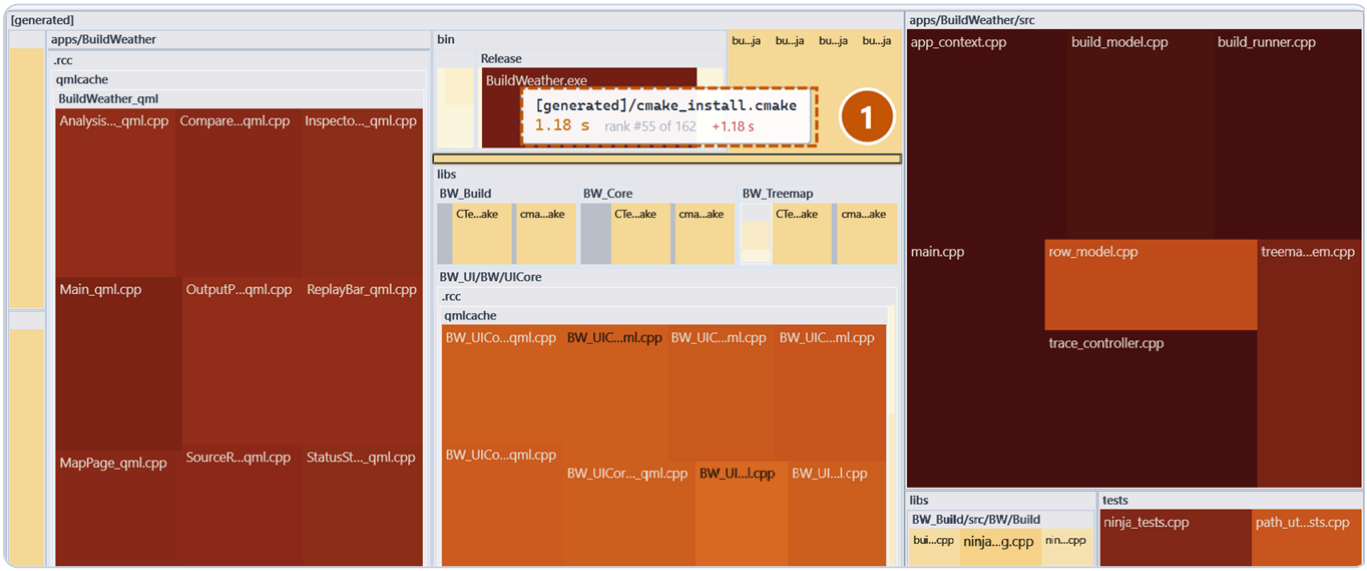
The three buckets

Not everything a build compiles is your code, and thousands of toolchain headers would swamp the picture if they were mixed in. Anything that is not a file under your source root is put in one of three synthetic directories:

BUCKET	WHAT LANDS THERE
`[generated]`	Anything under the build directory: moc output, qmlcache, protobuf, the linked binaries
`[external]`	Absolute paths outside both roots, such as a vendored SDK
`[system]`	Toolchain and SDK headers

4. Finding your way around the map

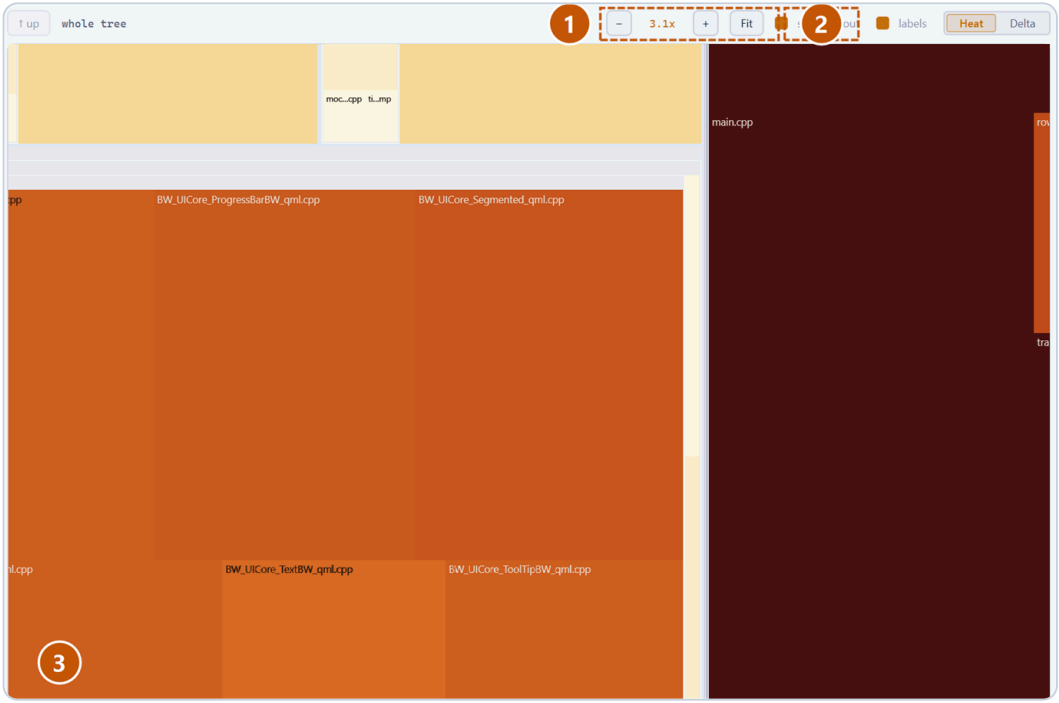
Hover any cell for its full path, its duration, and its rank.



Hovering a cell shows path, duration and rank

1. The tooltip follows the cell rather than the cursor, so it does not jitter while you scan across small files. Hovering a *directory* header gives you the total for everything inside it, which is the number you want when looking for a hot module.

On a real project the whole tree does not fit legibly on one screen. Scroll to zoom and drag to pan.



Zoomed in, with file names legible

1. Zoom out, the current factor, zoom in.
2. **Fit** goes back to the whole tree. `Ctrl+0` does the same.

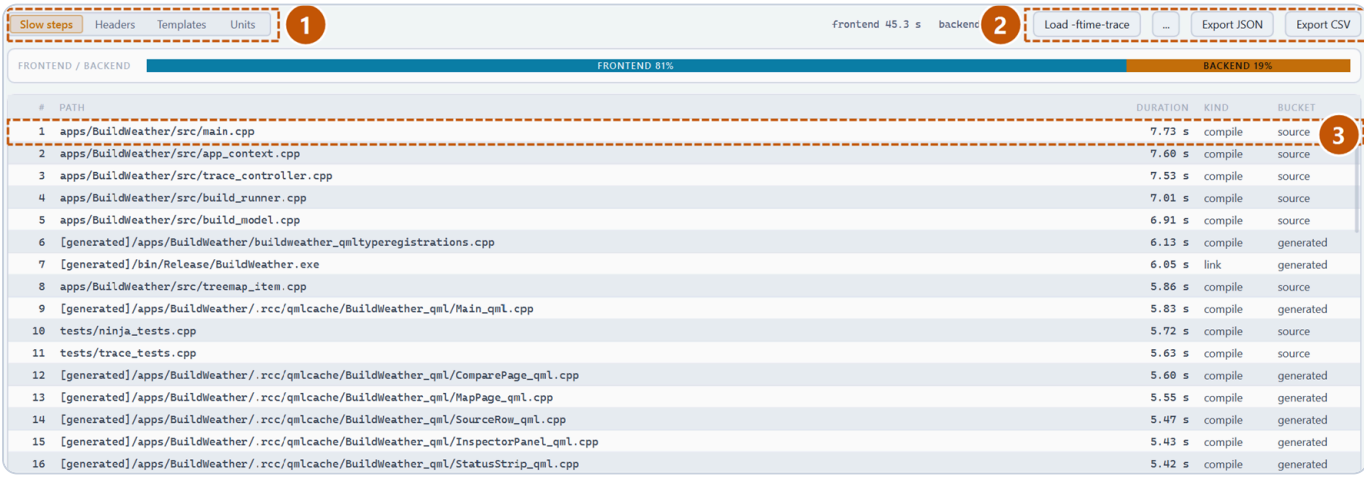
3. Zooming recomputes the layout rather than magnifying the picture, so cell labels stay sharp and more of them appear as there becomes room. This is how you read file names on a large project.

ACTION	GESTURE
Zoom about the cursor	Mouse wheel
Pan	Drag with the left button
Select a cell	Click
Drill into a directory	Double click
Back up one level	Right click, or ↑ **up**
Fit the whole tree	**Fit**, or `Ctrl+0`

Stable layout is on by default. It orders cells by name rather than by duration, so a file keeps its position from one build to the next and only its area changes. That is what makes two builds comparable by eye. Turning it off packs the rectangles into squarer shapes, at the cost of everything moving whenever a timing moves.

5. Finding the expensive step

The Analysis tab is deliberately plain: no animation, just numbers.



The slowest build steps

1. Four views: **Slow steps** from `.ninja_log`, then **Headers**, **Templates** and **Units**, which need `-ftime-trace` data.
2. **Export JSON** or **Export CSV** writes the current ranking to a file so you can paste numbers into a discussion or diff them in a script.
3. Each row is one step: rank, path, duration, what kind of step it was, and which of the three buckets it belongs to.

6. Finding the expensive header

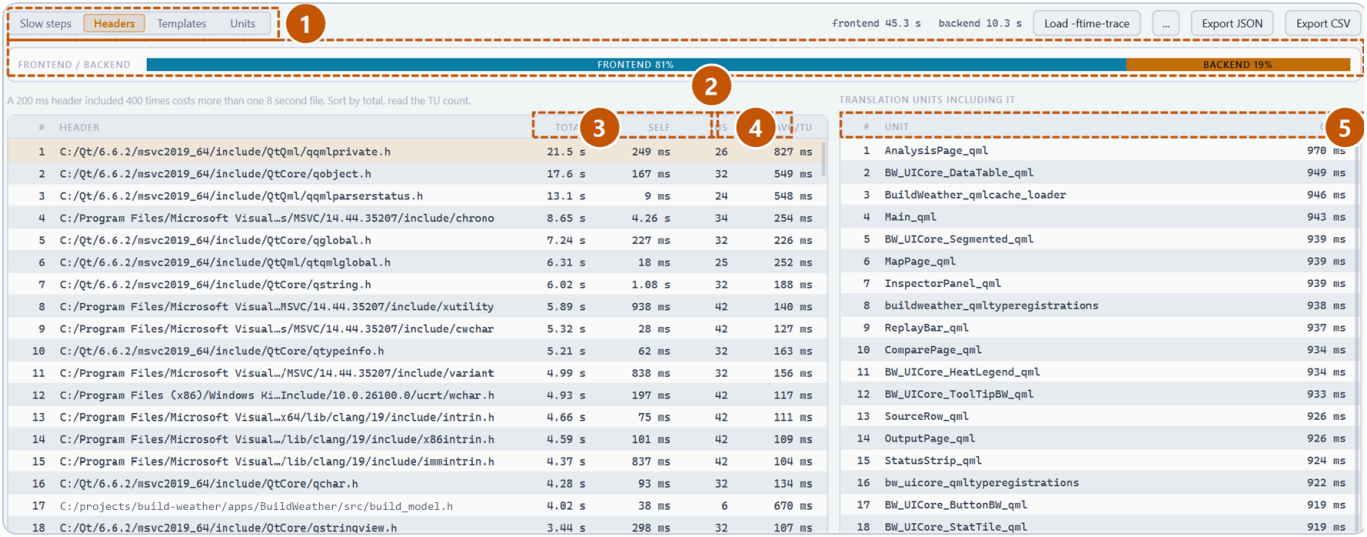
This is the view that pays for the tool, and it needs a clang-cl build with `-ftime-trace`. MSVC has no equivalent, so this is a second build directory used only to produce the data. Install Visual Studio's "C++ Clang tools for Windows" component, then:

```
cmake -S . -B build\clangcl -G "Visual Studio 17 2022" -A x64 -T ClangCL ^
      -DCMAKE_CXX_FLAGS="/DWIN32 /D_WINDOWS /EHsc /clang:-ftime-trace"
cmake --build build\clangcl --config Release
```

`-T` selects a toolset, which the Ninja generator does not support, hence the Visual Studio generator. And `CMAKE_CXX_FLAGS` replaces CMake's defaults rather than adding to them, which is why `/EHsc` has to be repeated: without it every `try` block fails to compile.

Then press **Load -ftime-trace** and point it at that directory.

For Build Weather's own tree, `tools\make-time-traces.cmd` does all of that into `build\clangcl-x64`.



Headers ranked by total cost across every translation unit

1. The **Headers** view.
2. The frontend / backend split for the whole project. A build that is mostly frontend is dominated by parsing and template instantiation, which is what include hygiene fixes. A build that is mostly backend is dominated by optimisation, which it does not.
3. **Total** is the cost of this header summed across every file that includes it. **Self** excludes what it pulls in, so a large total with a small self means the header is expensive because of *its* includes, not its own text.
4. **TUs** is how many translation units include it. This is the column that matters: a 200 ms header included 400 times costs more than one eight second file, and only this ranking shows that.
5. Click any header to list the translation units that include it, so you know where to start cutting.

7. Comparing two builds

Copy a `.ninja_log` aside before the change you want to measure, then load it as a baseline.

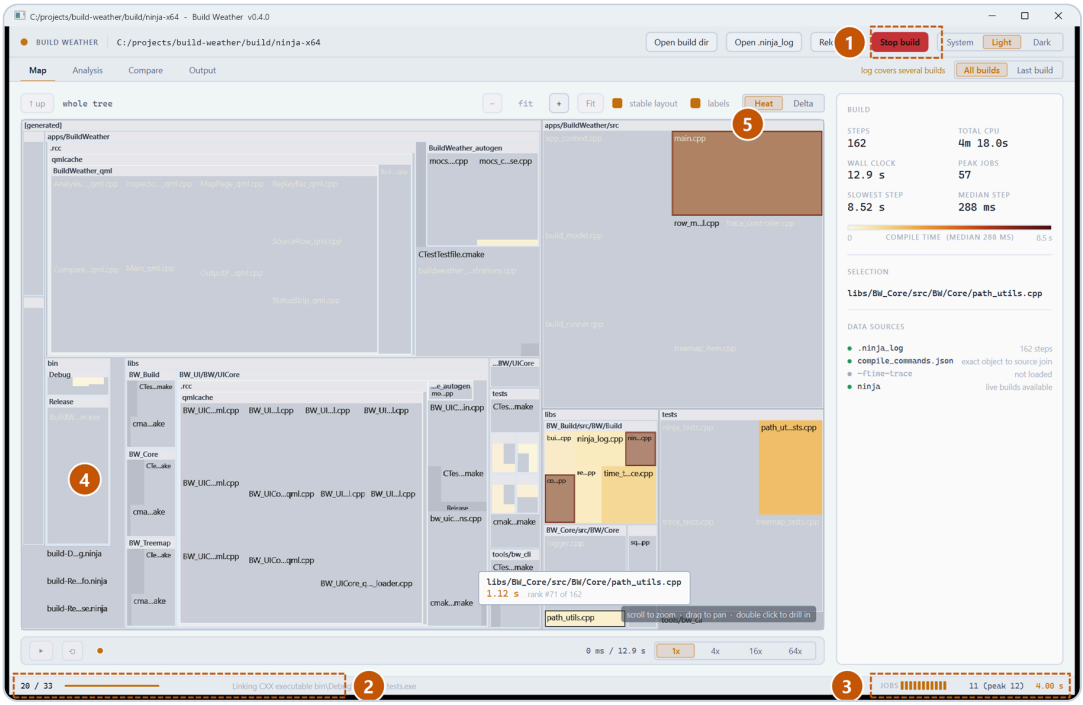
Per-file deltas against a baseline build

1. **Load baseline .ninja_log**, or pass `--baseline <file>` on the command line. **Clear** drops it again.
2. Baseline total, current total, the difference, and how many steps changed.
3. One row per step, sorted worst regression first. Red is slower, green is faster, and steps that only exist on one side are marked `added` or `removed`.

Switch the map to **Delta** colouring to see *where* in the tree the regression lives rather than just which files moved.

8. Watching a build

Press **Build** (or `Ctrl+B`) to run ninja in the loaded build directory.



A build in progress

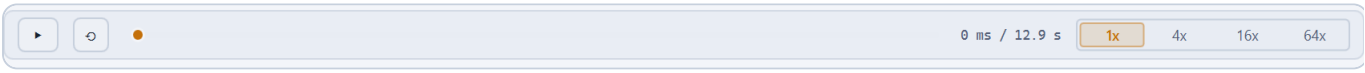
1. **Stop build** cancels it. The pip beside the wordmark beats while a build is running.
2. Progress through the plan and elapsed time.
3. **Jobs**: how many compiler processes are actually running right now, taken from ninja itself. Watching this collapse near the end of a build, because everything is waiting on one enormous translation unit, is genuinely diagnostic.
4. Cool grey cells have not been rebuilt. This build only recompiled part of the tree, and the grey is everything ninja decided was already up to date.
5. Cells flash as they complete and settle to their final colour over about 400 ms. With a dozen jobs in flight this reads as a shimmer crossing the map.

> Ninja only reports a step as it *finishes* when its output is a pipe, which is what a child process gets. So the map animates on completion rather than on start, and the job count comes from ninja's own counter rather than being inferred. The timings are exact either way.

The **Output** tab has the raw ninja output, with errors and warnings highlighted. When a build fails, that is where the reason is.

9. Replaying a build

You do not have to be there when a build runs. Every `.ninja_log` already contains a start and an end for each step, so any build already in the log can be replayed without having captured anything extra.



The replay transport

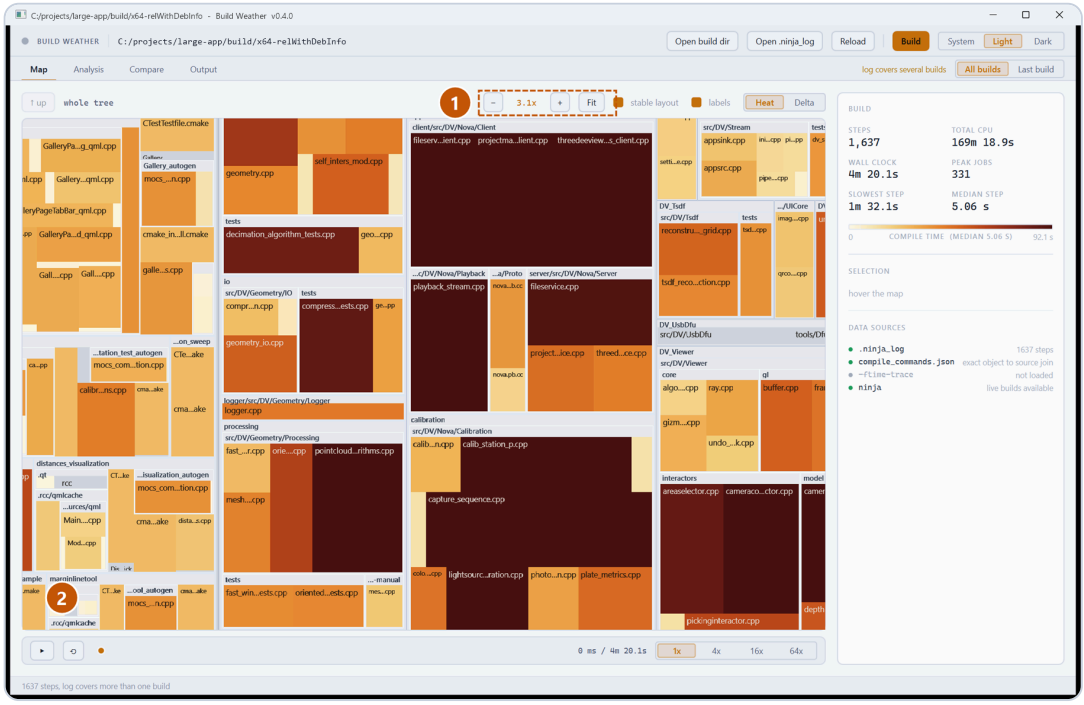
From left to right: **play / pause**, **rewind to the start**, the **scrubber**, the position and total, and the **playback rate**. An **Exit replay** button appears beside them once a replay is running, and `Esc` does the same.

Drag the scrubber to jump to any moment; the map shows exactly which files were in flight then, and the jobs readout in the status bar shows how many. That is the quickest way to answer "what was the build waiting on at the 40 second mark".

Two things to know:

- **Speed only has an effect while playing.** A four minute build at 1x takes four minutes to watch, so pick 16x or 64x. Choosing a rate starts playback for exactly this reason.
- **Switch the scope to Last build first.** Timestamps from different ninja runs come from different clocks, so replaying a log that covers several builds mixes them. The transport warns you when that is the case, and *Last build* narrows the data to a single run where the timeline is exact.

10. Large projects



A 1600-step project, zoomed in

1. Zoom is what makes a project of this size readable.
2. At fit zoom you can see the shape of the build and which modules are hot. Zoom in to read names, drill into a directory to give it the whole window.

Layout is recomputed on every zoom step and stays inside a frame at several thousand files. Cells scrolled out of view are not drawn at all.

11. Working from the command line

`bw_cli` runs the same analysis without a window, for CI or a quick answer.

```
bw_cli.exe analyze build\ninja --traces build\clangcl --top 20
```

```
bw_cli.exe compare old.ninja_log build\ninja\.ninja_log --csv deltas.csv
```

`analyze` prints the build summary, the slowest steps and the header ranking. `--json` writes the same data to a file, truncated to `--top` rows like the console output; `--csv` writes every row, never truncated.

`analyze` takes the build **directory**, not the log file. It finds the source root from `compile_commands.json`, so `--source` is only needed when there is no compile database.

Exit codes are worth knowing if you script this: **0** success, **1** an input could not be read or an output could not be written, **2** the command line was wrong. A flag given without its value is a command-line error rather than a silent no-op, so a mistyped `--csv` fails the step instead of quietly writing nothing.

`bw_cli --help` lists every option, and `bw_cli --version` prints the version.

12. What the numbers mean

Worth being precise about, because a treemap invites more trust than it earns.

Durations are exact. They come from `.ninja_log`, which records the start and end of every step in milliseconds. Nothing is sampled or estimated.

A log accumulates across builds. See the scope switch in section 2. When the header says "log covers several builds", the durations are still each exact but they were measured at different times, and the *timeline* mixes clocks. In that state the **wall clock and peak jobs figures mean little**, because they come from overlaying separate runs; switch the scope to *Last build* to get numbers from a single timeline. `bw_cli` prints the same warning in that case.

Several steps can share one file. A generated source is written by one step and compiled by another. Both belong at that leaf, so their durations are summed: that is what the file costs you.

Zero-duration steps are still drawn. They get a minimum area, so a fast file does not silently vanish from the tree.

`-ftime-trace` numbers describe a clang build. If your real build is MSVC, the absolute times will differ. The *ranking* of headers is what transfers.

13. Keyboard reference

KEY	ACTION
`Ctrl+O`	Open a build directory
`Ctrl+B`	Start or stop a build
`F5`	Re-read the log
`Ctrl+0`	Fit the map
`Ctrl++` / `Ctrl+-`	Zoom in / out
`Esc`	Leave replay